

HAT-P-55b

R-2179

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-55_260705 · created 2026-08-13T01:16:34+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461226.733 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.048
Transit depth (Rp/Rs)^2	2.57 +/- 1.54 [%]
Orbital Inclination (inc)	89.28 +/- 2.39
Airmass coefficient 1 (a1)	6.72 +/- 3.71
Airmass coefficient 2 (a2)	-1.99 +/- 1.66
Scatter in the residuals of the lightcurve fit is	57.31 %
Best Comparison Star	#1 - [517, 117]
Optimal Aperture	1.94
Optimal Annulus	5.75
Transit Duration (day)	0.108 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.16 $\pm$ 0.048 vs 0.1202 (deviation 0.83 σ)
Duration fit vs published	0.108 d vs 0.1225 d (deviation 1.04 σ)
Mid-time O–C	24.1 min
Depth SNR	3.3
Residual scatter	57.31 %

Light curve

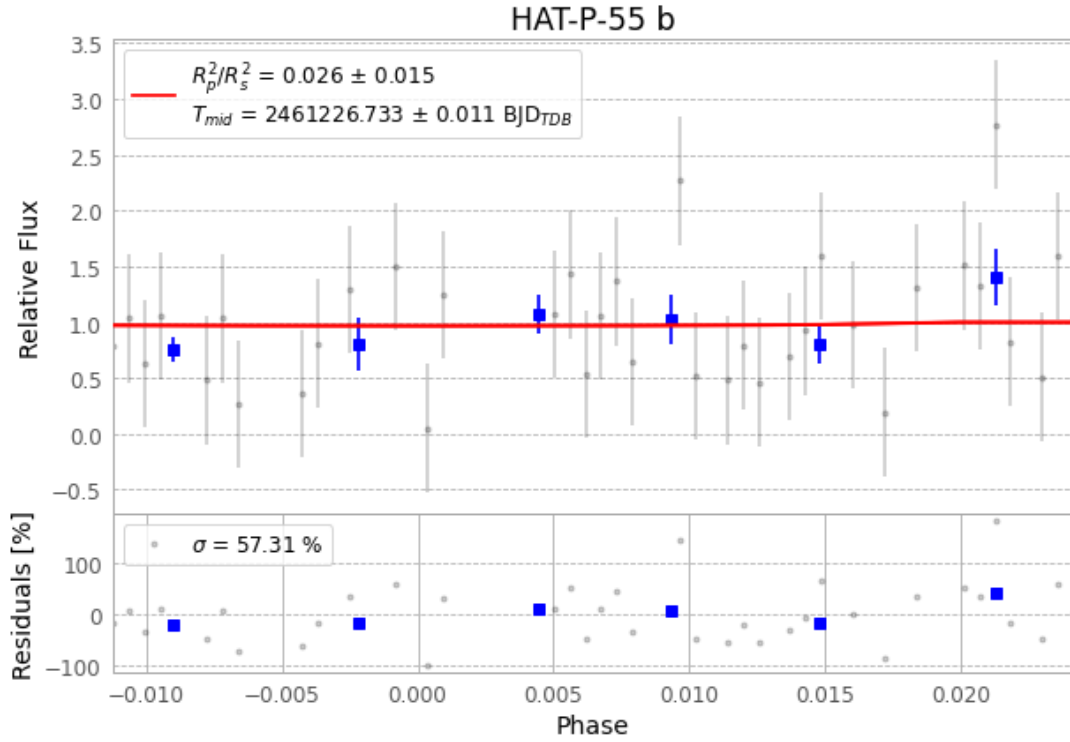


Figure 1 — FinalLightCurve_HAT-P-55 b_2026-07-04.png (SHA-256 7bf5cd7005c20ab0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [581, 415], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9f76818418e2d7a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

158 FITS frames (62 MB), HATP-55260705025519.FITS.gz → HATP-55260705073316.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-55-260705.log (fb2e33441b05...), FinalLightCurve_HAT-P-55 b_2026-07-04.png (7bf5cd7005c2...), FinalLightCurve_HAT-P-55 b_2026-07-04.csv (57c1a256d135...)

Compute resources

Wall time 123.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 01:16Z.